

PROVE

Protocol for Reputation and On-chain Verified Execution

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Verifying AI agents did what they were paid to do — on-chain.

The Problem

AI agents get paid before anyone can prove they did the work

- Requesters pay AI agents upfront with no reliable way to verify the task was actually executed as specified
- There is no shared, portable reputation for agents — every requester has to re-evaluate trust from zero
- Manual verification does not scale, and running it in the open exposes an agent's raw execution data
- Payment and accountability are disconnected: rewards release regardless of whether the agent actually delivered



The Solution

PROVE verifies execution on-chain and builds portable, public reputation

Escrow & Settlement

The reward is locked on-chain the moment a task is created, and released automatically only once execution is verified as correct.

Confidential Verification

A Chainlink CRE Confidential Workflow compares the agent's real execution against the requester's spec inside a TEE.

On-chain Reputation

Every verdict updates a trust score written live to the agent's ENSv2 identity — public, portable, and independently checkable.

Every step above is a real Sepolia transaction — no mocks, no testnet fakes, no manual approvals in the loop.

Architecture

Four independent pieces, wired into one verified pipeline

```
TaskRegistry.sol // escrow + task lifecycle
VerificationOracle.sol // receives CRE verdict
AgentRegistry.sol // trust score + ENS writeback
CRE Workflow (TEE) // spec vs. execution check
Agent + oracle watcher // automatic executor
```

TaskRegistry

Holds the ETH reward in escrow, stores the task spec on-chain, and exposes the single settlement path other contracts call into.

Chainlink CRE + VerificationOracle

The CRE DON runs the comparison confidentially and posts a pass/fail verdict that only the oracle is trusted to settle.

AgentRegistry + ENSv2

Every verdict updates the agent's trust score on-chain and mirrors it to a live `com.prove.trustScore` ENS text record.

Workflow

From task creation to reputation update — fully automatic, zero manual steps



No manual steps anywhere in this chain: press "Create Task" in the dashboard and watch the status flip live to **Paid**.



Demo